Project proposals

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Introduction

Goal of the projects of to apply the methods and tools studied in this module to a (practical) problem, and/or to study alternative methods in the area of automated planning.

Projects can be done in group (max 3-4 people), and may be the same groups already defined for the part about Intelligent Agents, or may be different (also individuale projects are fine).

There are two types of projects this year: a **didactic** project and **research** projects.

Project types

In the didactic project, you are given a description of a problem (of similar difficulty of those seen in class), you have to make the PDDL+ encoding, and run the model on some instances. Details of the possible research projests are given below.

The presentation of the **didactic** project will be part of the exam, which will be completed by an oral exam (in the same day) to define the final grade for the Automated Planning module (6CFU).

Research projects are completed by:

- a report, of max 5 pg in a format and with a given structure I ll make available (to be sent 5 days before the actual date of the exam);
- a presentation, done to me and possibly to other researchers that proposed the project and will help me monitoring. If done in group, I expect that each member of a group will be presenting one part of the project.

Project+report+presentation will define the final grade for the Automated Planning module (6CFU).

Project #1: Classical and Numeric planning

There are some (video game) problems for which a classical PDDL encoding is given, we would like to design an encoding in numeric planning for it, and evaluate the relative performance of the two encodings.

The papers can be found also in Teams, and refer to Puzznic and Plotting: A planning problem with Complex Transition.

Project #2: Map Construction in PDDL+

Implement a software tool that extracts from openstreetmap city maps and encode such a maps in PDDL+.

The are already some APIs and intermediate tools that support the extraction and simplification of the maps, and thus the focus would be on the encoding in PDDL+.

Project #3: Train timetabling

We are in contact with Trenitalia, and we got some (pubblicly available) dataset about the train's timetable in Sardinia and Lombardia.

The idea of this project is to use ASP for the verification of already implemented timetables, or the construction of new timetables.

Project #4: Axioms in PDDL

Axioms are PDDL constructs that we did not see in class: they express logical consequences that are automatically inferred from the current state of the world, rather than being explicitly changed by an action's effects.

While quite useful, they are not supported by all solvers, given that they are not handled efficiency by PDDL solvers since such implicit knowledge would be more amenable to be (represented and) solverd by a constraint solvers.

In this context, it would be possible to do two projects:

- incorporate such axioms directly in the planner,
- let the axioms to be evaluated by an external solver, e.g., an ASP solver, that interact with the planner.

Project #5: Planning and Answer Set Programming

During the last lecture we have seen another approach to planning, i.e. planning via Answer Set Programming.

The idea of this project is to consider planning/scheduling problems modeled and solved with Answer Set Programming. Possible projects refer to:

- model a new problem in ASP, in particular scheduling of night pharmacies;
- extend/improve an existing ASP encoding;
- improve the solving part with, e.g., domain heuristcs.

Most of the examples are in the Healthcare domain: Chemotherapy Treament, Rehabilitation Session, Nurse Scheduling, Operating Room Scheduling, Nuclear Scheduling ... for which we have data from various Hospitals (ASL1 Liguria, S. Martino, Medipass, Annunziata).

Project #6: Integration of Machine Learning and Planning

This type of projects is somehow linked to the previous. In project #5, with the real data it is basically possible to (*i*) test whether the ASP solution can simulate the real data, i.e., the ASP solution can compute the same schedule(s) the Hospital did, and (*ii*) propose alternative schedules that could have been performed, better than the one scheduled by the hospital according to some parameters (OR usage, patient's waiting time, number of patient served, use of resources, ...).

However, it is not possible to define *provisional* schedules.

Starting from one of the problem, and corresponding, data, the goal of this project would be to:

- estimate the values of some of the main feature(s) of the problem (e.g., surgery duration in ORS) via ML algorithms;
- use these predictions and related statistics (confidence) to update the ASP encoding in order to produce "better" schedules (e.g., confidence to compute better schedule in terms of OR usage in ORS).